

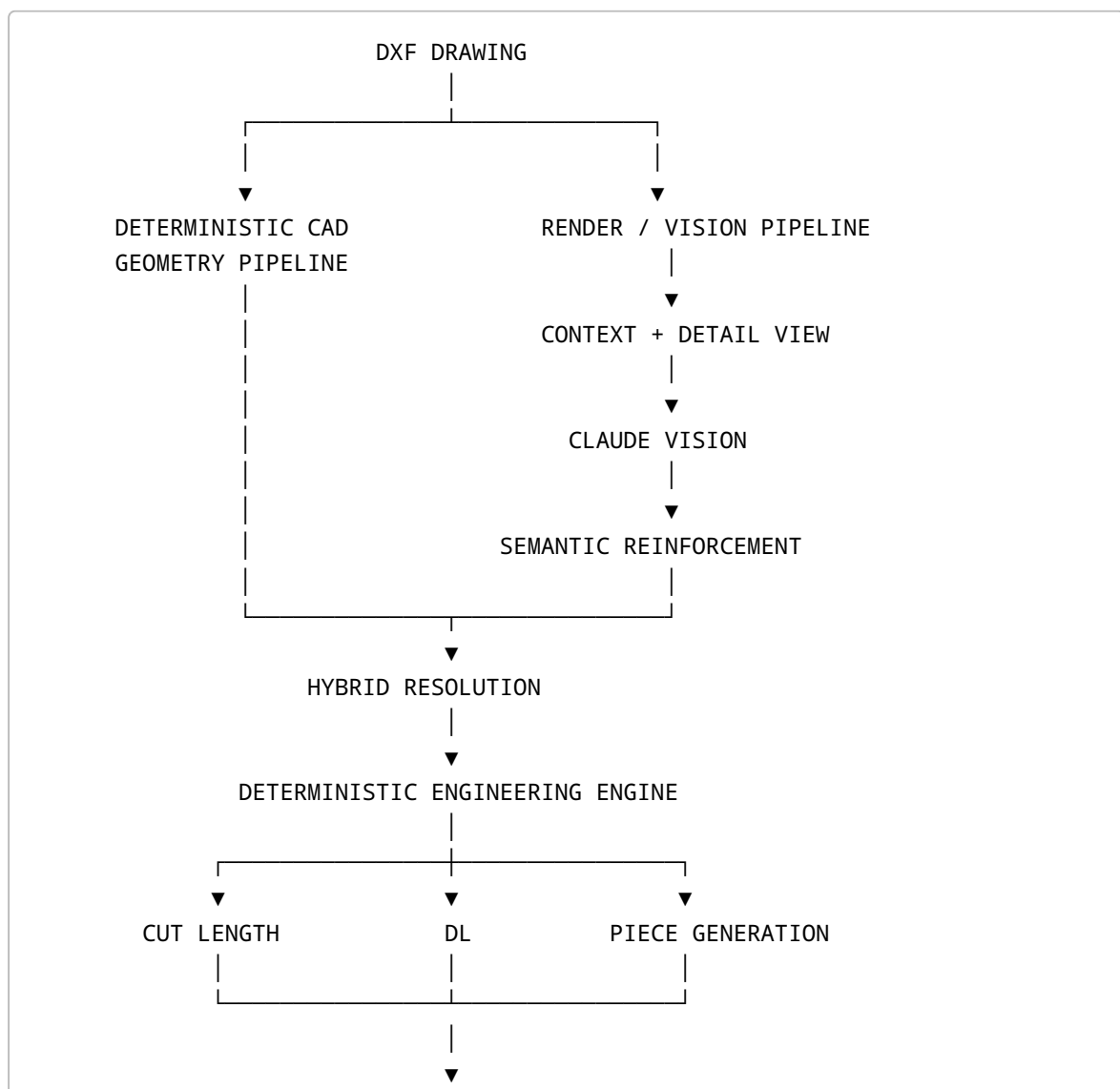
Architecture Summary — Hybrid Steel Beam Estimation System

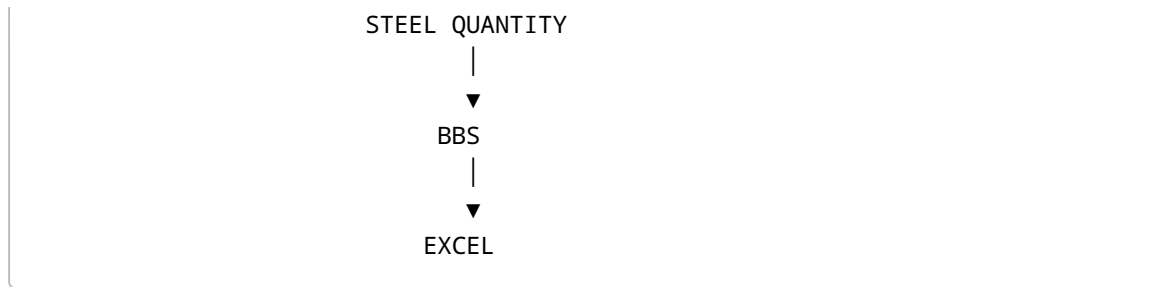
1. System Architecture

The Steel Beam Estimation system is a **hybrid engineering intelligence system** combining deterministic CAD/engineering computation with AI-based visual reinforcement interpretation.

The architecture is designed around a strict separation of responsibilities:

- **AI Vision** interprets reinforcement information that is difficult to reliably derive from raw CAD structure alone.
- **Deterministic engineering logic** remains authoritative for geometry, engineering rules, calculations, quantities and final estimation.
- The two outputs are combined through a **Hybrid Resolution layer** before final quantity computation.





2. Vision Responsibility

Claude Vision is primarily responsible for **semantic perception and reinforcement interpretation**.

Vision is preferred for identifying:

- Reinforcement target
- Drawing layer/context
- Physical reinforcement groups
- Bar count
- Bar diameter
- Reinforcement specification
- MAIN / EXTRA classification
- Support scope
- Visual stirrup identification
- Relationships between reinforcement annotations and the represented beam/detail

Vision therefore answers the question:

"What reinforcement exists in this drawing and what does it represent?"

3. Deterministic Engineering Authority

The deterministic engineering engine remains the authority for engineering and quantity computation.

It is responsible for:

- CAD geometry
- Beam geometry
- Spacers and cover
- Clear span / engineering length
- Bar geometry
- Cut length
- Development length
- Anchorage
- Hooks and bends
- Stirrup quantity and engineering
- Piece generation
- Unit weight

- Steel quantity
- BBS generation
- Excel output
- Engineering validation

The deterministic engine therefore answers:

"How should the identified reinforcement be engineered and quantified?"

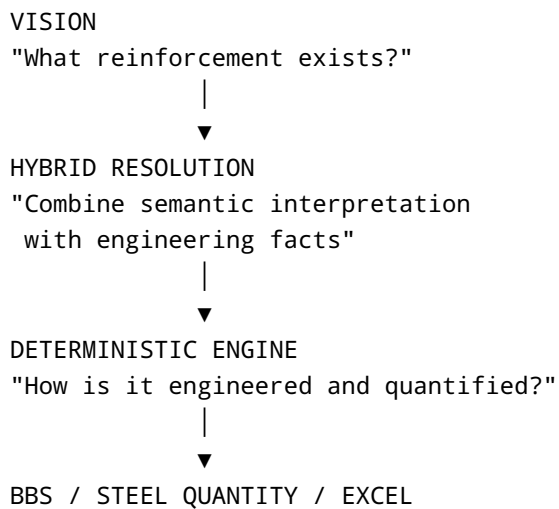
4. Hybrid Resolution

The Hybrid Resolution layer combines:

1. Deterministic geometric facts extracted from the DXF.
2. Semantic reinforcement interpretation produced by Vision.
3. Project engineering rules and estimator practices.

This prevents the AI model from becoming the final calculation authority.

The architecture deliberately follows:



5. Core Design Principle

The fundamental architecture principle is:

Vision decides what reinforcement exists, while the deterministic engineering engine decides how that reinforcement is engineered and quantified.

This separation provides the flexibility of AI-based drawing understanding while preserving deterministic, auditable engineering calculations.